RAKSHAM · ROUND 1 · TRACK 03

TRUVY

Know what's real.

AI-powered trust intelligence for financial communications.

TEAM RED HAT

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Challenge · Deepfake Detection for Financial Communications



A Familiar Voice Is No Longer Proof.

ILLUSTRATIVE SCENARIO · NOT A REAL CASE

A finance officer receives a voice note that sounds exactly like the company's CEO.

The message requests an urgent transfer to a new beneficiary. A convincing video appears to confirm the instruction. The sender discourages a callback, claiming to be unavailable.

But the real executive never authorised the transaction.

01 · FAMILIARITY

Familiarity creates trust.

Employees naturally recognise voices, faces and communication patterns. Attackers exploit this familiarity.

02 · URGENCY

Urgency suppresses verification.

Time pressure, authority claims, secrecy and callback avoidance encourage employees to bypass normal controls.

03 · DETECTION

Detection does not equal prevention.

A media detector can surface signs of manipulation, but its score alone cannot establish whether the financial instruction is legitimate.

HARM

- Fraudulent transfers
- Unauthorised beneficiary changes
- Difficult recovery
- Operational & reputational damage

The vulnerability is not simply a fake voice. It is the gap between recognising someone and verifying that they authorised an instruction.

From Detecting Deception to Verifying Intent.

01

DETECT

Identify manipulation

Analyse audio and video for signs of synthetic media or manipulation.

02

ASSESS

Weigh the context

Combine forensic findings with language confidence, pressure cues and payment context.

03

VERIFY

Route to trust

Recommend trusted-channel callbacks, human review and evidence documentation.

DIFFERENTIATION

Conventional media analysis focuses on whether content **appears manipulated**. TRUVY connects media analysis to an **operational decision** and a **verification process** · it does not rely on a single detector score.

TRUVY helps answer not only whether media looks suspicious, but what should happen before the associated payment proceeds.

Every Suspicious Instruction Has a Verification Path.

Incoming instruction
Audio or video · payment or beneficiary change
INTAKE

↓

MEDIA ANALYSIS
- Audio forensics
- Video analysis
- Language identification
- Provenance checks where available

CONTEXT ANALYSIS
- Pressure cues
- Transaction amount
- Beneficiary changes
- Timing & communication context

↓

Risk assessment
Combine signals under explicit rules · a risk band

↓

LOW
Continue
Proceed through normal maker-checker controls and log the case.

ELEVATED
Callback
Require an independent callback before payment release.

HIGH
Hold & review
Hold the payment for human review and require dual approval.

EVIDENCE Every outcome writes an **evidence record and report package**.

Critical rule: callback numbers come from the institution's trusted directory, never from the suspicious message. TRUVY does not automatically reject payments.

Multiple Signals. One Explainable Risk Assessment.

LAYER 01

Media forensics

Audio deepfake detection, sampled-frame video analysis, language identification and provenance checks.

- Audio · Video · Language · Provenance

LAYER 02

Communication analysis

Speech-to-text and structured identification of urgency, secrecy, authority claims and callback avoidance.

- Transcript · Pressure cues

LAYER 03

Financial context

Transaction amount, beneficiary history, unusual timing and communication-channel anomalies.

- Amount · Beneficiary · Timing

LAYER 04

Decision engine

Combine signals, apply explicit rules, generate an explanation and route the request for the appropriate action.

- Rules · Explanation · Routing

EXAMPLE RULE

A new or changed beneficiary above a configured threshold, combined with a pressure cue, triggers a callback **regardless of the audio detector's result**.

LOW	ELEVATED	HIGH
Normal controls.	Trusted callback.	Hold + human review.

The detector informs the decision. It does not control the decision alone.

Measure the Risk. Validate the Decisions.

VALIDATION PLAN · RESULTS PENDING

No measured results are claimed at this stage.

EQUAL ERROR RATE

•

PENDING

FALSE-POSITIVE RATE

•

PENDING

PER-LANGUAGE

•

PENDING

END-TO-END LATENCY

•

PENDING

CORRECT ESCALATIONS

•

PENDING

LANGUAGE	EER	FPR	STATUS
English	·	·	Pending
Hindi	·	·	Pending
Tamil	·	·	Pending

RESOURCES UNDER EVALUATION

SEA-Spoof · HAV-DF · IndieFake · IndicSynth · SafeEar

Language performance

Evaluate English, Hindi and Tamil samples.

Recording conditions

Compare clean audio with compressed recordings representative of messaging apps.

Decision quality

Genuine instructions, impersonation scenarios, and cases where context matters despite an inconclusive detector.

Operational performance

Measure processing latency, false-positive rates and time to resolution.

Trust Requires More Than Detection.

PRIVACY

- Prototype uses synthetic, public or consented data
- No customer credentials
- No voiceprint database
- Ephemeral media processing where feasible
- Defined retention for escalated evidence

SECURITY

- Role-based access
- Encryption in transit and at rest
- Tamper-evident audit logs
- Rate limiting
- Deployment within the institution's security boundary

HUMAN CONTROL

- No automatic payment rejection
- No fraud accusation from one weak signal
- Human review for high-risk cases
- Documented reviewer overrides
- Dual approval where required

False positive

Verification or review introduces friction, but TRUVY does not independently reject the payment.

False negative

Independent payment-context rules may still trigger verification even when the media detector fails.

TRUVY is designed to support safer decisions even when its media detector is imperfect.

Built to Fit Existing Financial Workflows.

MAKER

Prepares instruction

TRUVY

Verification gate

RECOMMENDED ACTION

- Normal controls
- Trusted callback
- Human review

CHECKER / APPROVER

Authorises release

ADOPTION PATHWAY

STAGE 01

Shadow mode

Measure recommendations without imposing additional transaction friction.

STAGE 02

Advisory mode

Present explanations and recommended actions to authorised approvers.

STAGE 03

Controlled enforcement

Introduce required verification for designated risk categories with human oversight.

SCALABILITY

- Separate media-analysis services from decision logic
- Support asynchronous processing
- Allow detectors to be replaced
- Add communication modalities through modular components

Conceptual architecture · not a claim of an existing production integration.

From Signal to Decision in 48 Hours.

48

HOURS · SIX CHECKPOINTS · ONE END-TO-END DEMONSTRATOR

00 · 06	06 · 16	16 · 28	28 · 38	38 · 45	45 · 48
Environment, data preparation, baseline detector.	Audio pipeline, language ID, transcription, video analysis.	Payment-context rules, decision engine, explanations.	Evaluation and language-aware calibration.	Callback workflow, reviewer queue, evidence package.	Fallbacks, limitations and rehearsal.
One clip produces a score.	Multiple signals for a staged scenario.	Three scenarios produce bands & reasons.	Initial validation tables.	End-to-end demonstration.	Clean final run-through.

DEMO SCENARIOS

01
Legitimate routine instruction.

02
Suspected deepfake requesting a transfer to a new beneficiary.

03
Low-confidence language scenario requiring contextual verification.

FALLBACK

If the media detector underperforms, retain the contextual risk and verification workflow, present the detector as advisory, and report the measured limitations.

CLOSING

A Familiar Voice Shouldn't Be Enough to Move Money.

TRUVY combines media analysis, contextual risk assessment, trusted verification and evidence-backed human review to help financial institutions make safer decisions before payment release.

DETECT

Identify potential media manipulation.

ASSESS

Evaluate communication signals and financial context.

VERIFY

Route suspicious instructions through trusted channels.

DOCUMENT

Preserve reasons and evidence for human review.

TRUVY

Know what's real.

TEAM RED HAT

RAKSHAM · TRACK 03

From Financial Communication to Verifiable Decision.

TRUVY combines media forensics, contextual risk analysis, and independent verification into a human-supervised workflow.

01 | INCOMING COMMUNICATION

Audio message
Voice note or recording

Video message
Video recording or extract

Metadata and transaction context
Sender, channel, timing, amount, beneficiary

02 | SECURE INGESTION

Input validation
Format, size, rate limiting

Secure media processing
Normalization, decoding, temporary artifacts

Provenance and evidence
Hashes, timestamps, C2PA where available

UNTRUSTED INPUT · CONTROLLED PROCESSING

Normalized media + metadata + evidence reference

03 | PARALLEL SIGNAL EXTRACTION

AUDIO FORENSICS
- Audio authenticity model
- Quality and manipulation indicators

Audio evidence + confidence

VIDEO FORENSICS
- Frame sampling and aggregation
- Video authenticity signals

Video evidence + confidence

LANGUAGE AND SPEECH
- Language ID, STT, confidence
- Structured pressure cues

Language and context signals

PROVENANCE AND PAYMENT
- C2PA and provenance
- Beneficiary, timing, amount, policy

Context and provenance evidence

04 | EVIDENCE FUSION AND POLICY ENGINE

- Signal availability and quality gate
- Evidence fusion: authenticity, context, provenance
- Policy rules and explainable rationale
- Risk classification and structured assessment

05 | CONTROLLED RESPONSE

- LOW — Normal maker-checker controls and audit logging
- ELEVATED — Independent callback through a trusted directory
- HIGH — Hold for human review and required approval

Risk determines verification requirements, not a verdict of fraud.

06 | HUMAN-LED FINANCIAL WORKFLOW

Authorized reviewer
Inspect signals, rationale, and evidence

Independent verification
Use trusted institutional contact details

Maker-checker workflow
Institutional approval controls retained

AUDIT TRAIL
Case IDs, hashes, timestamps, versions, recorded decisions

Multiple Signals. One Explainable Risk Assessment.

TRUVY separates media authenticity, contextual warning signs, and institutional policy before recommending an action.

INPUTS
Media + Context

NORMALIZATION
Secure preprocessing
Decode • quality checks

PARALLEL SIGNAL EXTRACTION

AUDIO	VIDEO	LANGUAGE & SPEECH	PROVENANCE & PAYMENT
Preprocessing • authenticity	Frame sampling • aggregation	Lang ID • STT • pressure cues	C2PA • beneficiary • timing • policy
Audio + confidence	Video + confidence	Structured cues	Context signals

RISK BAND	INTERPRETATION	RECOMMENDED ACTION	QUALIFICATION
LOW	No escalation threshold met	Normal maker-checker • log	Does not guarantee genuine
ELEVATED	Indicators justify verification	Independent callback (trusted directory)	Request for verification • not fraud
HIGH	Policy concern requiring escalation	Hold • human review • dual approval	Not automatic fraud finding

EVIDENCE PACKAGE

SIGNALS	CONTEXT	AUDIT
Scores, confidence, availability	Transcript, timing, beneficiary	Hashes, versions, rationale

INDEPENDENT VERIFICATION LOOP

1. **Trusted directory lookup**
2. **Independent callback**
3. **Authorized human decision**

Never use contact from suspicious message. Final decision remains institutional.

Trust Must Be Engineered, Not Assumed.

TRUVY limits data exposure, resists common tactics, and makes model limitations visible to reviewers.

SECURITY & PRIVACY LAYERS

- Auth • RBAC • least privilege
- Encryption in transit/at rest
- Tamper-evident audit
- Minimized retention
- Versioned policies/models

No voiceprint database. Hashes, not full raw exports by default.

VALIDATION (PLANNED)

- English/Hindi/Tamil
- Clean vs compressed
- EER/FPR per condition
- Latency/failure rates

PLANNED VALIDATION · RESULTS PENDING

THREAT	POTENTIAL FAILURE	MITIGATION	RESIDUAL
Synthetic media	Evasion / subtle edits	Multi-signal + independent verification + human review	Probabilistic; cannot guarantee detection
Low-quality/compressed	Unreliable scores	Quality gate marks signals unavailable/low conf	May remain inconclusive
Callback manipulation	Fake contact	Use trusted directory only	Depends on directory integrity
False pos/neg	Business friction or missed fraud	Verification, hold, review, monitoring	Errors cannot be eliminated

— THANK YOU

TRUVY

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AI, Model, Dataset and Asset Disclosures.

Supplementary disclosures covering AI assistance, technical components, datasets, and third-party assets.

SUPPLEMENTARY DISCLOSURE · SEPARATE FROM THE 10-SLIDE PITCH AND 3-SLIDE ARCHITECTURE

A · AI-ASSISTED CONTENT

MATERIAL	DISCLOSURE
Deck wording, structure and code	AI-assisted drafting and development using the tools identified by the team. Final wording and code should be reviewed by the team.
Architecture structure and diagrams	AI-assisted planning and HTML/CSS implementation. Diagrams are rendered using the presentation's HTML and CSS.
Branding and visual identity	Team-led product identity and design decisions. Confirm any AI-assisted or externally sourced elements before submission.

Technical disclosure: No measured detection performance is claimed. The proposed runtime design does not delegate payment decisions to a language model. Any planned language-model component is limited to producing structured pressure-cue indicators for downstream assessment.

B · MODELS AND COMPONENTS

COMPONENT	MODEL / APPROACH	STATUS
Media detection and signal extraction	Audio, video, speech-to-text and language identification	UNDER EVALUATION
Pressure-cue extraction	Structured rules or templates; optional model-assisted flags	PLANNED
Media provenance	C2PA-compatible verification, where available	UNDER EVALUATION
Risk and policy engine	Deterministic rules over available signals	PLANNED
Evidence and audit record	Evidence hashing and tamper-evident logging	PLANNED

No detection model is finally selected. Model selection depends on licence review, suitability, language coverage, and the organisers' applicable rules.

C · DATASET CANDIDATES

Not evidence of dataset usage

- **SEA-Spoof**
  Candidate reference; source and licence to verify.
- **HAV-DF**
  Candidate reference; source and licence to verify.
- **IndieFake**
  Candidate reference; source and licence to verify.
- **IndicSynth**
  Candidate reference; availability and terms to verify.
- **SafeEar**
  Reference only; dataset access and licence to verify.
- **Synthetic scenarios**
  Team-created scenarios, subject to review.

Dataset usage status: No dataset has been used in the prototype as of submission, if confirmed by the team. Candidate resources remain subject to access, licensing, and suitability review.

D · THIRD-PARTY ASSETS AND LICENCES

ASSET	SOURCE	LICENCE / STATUS
Manrope, Inter, IBM Plex Mono	Google Fonts or original font repositories	Verify each font's applicable licence.
Icons and diagrams	Inline SVG in the presentation	Team-authored assets, subject to verification.
TRUVY logo	Team-supplied asset	Confirm ownership and asset provenance.
C2PA-compatible tooling	Candidate tooling	Exact tool and licence require confirmation.
Reveal.js	Legacy dependency, if retained	Confirm whether it is included or used in this build.

Third-party sources and licences must be verified against the actual assets and dependencies included in the final submission.

Implementation, Evaluation and Key Limitations.

What exists today, how it will be measured, and the boundaries the team acknowledges.

A · IMPLEMENTATION STATUS

IMPLEMENTED · UNDER EVALUATION · PLANNED

This submission presents the proposed TRUVY design and architecture, not a running financial-security product. The current deliverables are the presentation and written design. No runtime detection, risk-decision, or callback workflow is included in this PDF.

The proposed product components are planned unless separately demonstrated and verified by the team. These include media analysis, payment-context assessment, pressure-cue extraction, evidence fusion, risk-band generation, trusted-directory callback verification, and evidence and audit records.

C · OPEN DEPENDENCIES

- Access, suitability, and licence review for candidate datasets and models.
- Confirmation of the organisers' rules for pretrained weights, third-party code, and external services.
- Institutional authorisation, security review, and operational controls before any real deployment.
- Access to an accurate institutional contact directory for independent callback verification.

B · EVALUATION STATUS

No measured result is claimed

DIMENSION	VALIDATION METHOD	STATUS
Language performance	Per-language EER on suitable licensed English, Hindi, and Tamil evaluation data, where available	PENDING
Genuine vs. manipulated	Bona fide and spoof comparison; false-positive rate	PENDING
Clean vs. compressed	Evaluate performance across recording and codec conditions	PENDING
False positives and negatives	Confusion analysis at defined decision thresholds	PENDING
Latency and failures	Timed processing, error handling, and fallback behaviour	PENDING
Policy and escalation correctness	Staged scenarios against expected risk bands and verification actions	PENDING

Metrics remain `PENDING` until evaluation is completed. No language coverage, detection accuracy, or latency is claimed without supporting evidence.

D · KEY LIMITATIONS

Boundaries of the proposed system

01 Detectors can miss unseen or evolving attacks.
02 Noise and compression can reduce detection reliability.
03 Performance may vary by language, accent, and recording conditions.
04 A detector score is not proof of fraud or authenticity.
05 Missing provenance metadata does not establish manipulation.
06 TRUVY does not automatically reject payments.